

Docker is Emerging as the Future of Application Delivery

This is a discussion on the role of Docker in software development and how it scores over virtual machines. As it becomes increasingly popular, let's look at what the future holds for Docker.



We all know that Docker is simple to get up and running on our local machines. But seamlessly transitioning our honed application stacks from development to production is problematic.

Docker Cloud makes it easy to provision nodes from existing cloud providers. If you already have an account with an Infrastructure-as-a-Service (IaaS) provider, you can provision new nodes directly from within Docker Cloud, which can play a crucial role in digital transformation.

For many hosting providers, the easiest way to deploy and manage containers is via Docker Machine drivers. Today we have native support for nine major cloud providers:

- Amazon Web Services
- Microsoft Azure
- Digital Ocean
- Exoscale
- Google Compute Engine
- OpenStack
- Rackspace
- IBM Softlayer
- Packet.net

AWS is the biggest cloud-hosting service on the planet and offers support for Docker across most of its standard EC2 machines. Google's container hosting and management service is underpinned by Kubernetes, its own open source project that powers many large container-based infrastructures. More are likely to follow soon, and you may be able to use the generic driver for other hosts.

Docker Cloud provides a hosted registry service with build and testing facilities for Dockerised application images, tools to help you set up and manage host infrastructure, and application life cycle features to automate deploying (and redeploying) services created from images. It also allows you to publish Dockerised images on the Internet either publicly or privately. Docker Cloud can also store pre-built images, or link to your source code so it can build the code into Docker images, and optionally test the resulting images before pushing them to a repository.

Virtual machines (VM) vs Docker

Some of the companies investing in Docker and containers are Google, Microsoft and IBM. But just because containers